

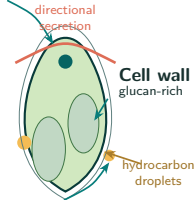
Architecture of a *Botryococcus braunii* colony

A three-scale cutaway of cell polarity, extracellular matrix, and biological partitioning

1 Polarized cell

Apical pole

retaining-wall / sheath precursors



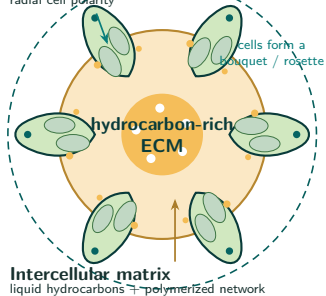
Basal-lateral pole

cortical ER-associated hydrocarbon export

2 Rosette / subcolony

Apical ends point outward

radial cell polarity



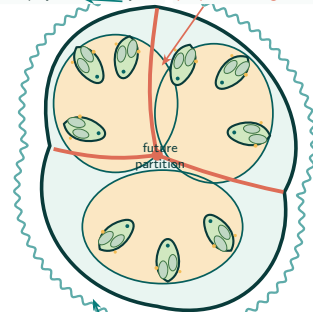
3 Whole-colony cutaway

Fibrillar sheath

outer polysaccharide layer

Zipper folds

pre-pattern daughter units



cell polarity

directional secretion

ECM architecture

cohesion & partitioning